

Introduction

- Learning rate is a configurable hyper parameter that controls the rate of how quickly the model is adapted to the problem i.e., determine the convergence rate of the model.
- High learning rate can leads to faster convergence but may also cause it to overshoot or oscillate around the optimal solution.
- Low learning rate can result in slow convergence and may stuck in suboptimal solution.
- Experimenting on finding a good learning rate for the model and dataset can be time consuming.

- Learning rate annealing is a technique used to adjust the learning rate during the training of the neural networks that helps the model to converge more efficiently and accurately.
- Common strategies for learning rate annealing are:
 - Time based decay
 - Step decay
 - Exponential decay
- These learning rate annealing method follows a predetermined schedule which have nothing to do with the performance of the model.

- Adaptive learning rate methods are an optimization of gradient descent method with the goal of minimizing the objective function of a network by using the gradient of the function and the parameter of the network.
- Adaptive learning rate automatically adjusts the learning rate based on the gradient information i.e., the learning rate changes dynamically in response to the actual training progress, leading to more efficient and effective learning.

- Gradient descent is an iterative optimization algorithm for finding the local minimum of a function.
- Gradient descent is the way to minimize objective function by updating model's parameter in the opposite direction of the gradient.
- Momentum, AdaGrad, RMSProp, ADAM are the adaptive learning rate strategies.

- Momentum: uses past gradient to smooth out the parameter update.
 - $v_t = \beta \cdot v_{t-1} + (1 - \beta) \cdot dw_t$
 - $w_{t+1} = w_t - \alpha \cdot v_t$
- AdaGrad: Adapts the learning rate for each parameter by dividing it by the square root of the sum of all its historical squared gradients.
 - $v_t = v_{t-1} + dw_t^2$
 - $w_{t+1} = w_t - \frac{\alpha}{\sqrt{v_t + \epsilon}} \cdot dw_t$

- RMSProp: keep the moving average of the squared gradients for each weight. And then divide the learning rate by root mean square of the squared gradients.

$$- v_t = \beta \cdot v_t + (1 - \beta) \cdot dw_t^2$$

$$- w_{t+1} = w_t - \frac{\alpha}{\sqrt{v_t + \epsilon}} \cdot dw_t$$

- Adam: combines the advantages of RMSProp and momentum by maintaining a moving average of both the gradient and its square.

- $v_t = \beta_1 \cdot v_{t-1} + (1 - \beta_1) \cdot dw_t$

- $s_t = \beta_2 \cdot s_{t-1} + (1 - \beta_2) \cdot dw_t^2$

Bias correction:

- $v_t = \frac{v_t}{(1 - \beta_1^t)}$

- $s_t = \frac{s_t}{(1 - \beta_2^t)}$

- $w_{t+1} = w_t - \frac{\alpha}{\sqrt{s_t + \epsilon}} \cdot v_t$

Problem Statement

- Among the different optimizers, ADAM is said to be the best optimizer for model performance but that is mainly in theory and need to be analyzed. This analysis is mainly focused on showing that ADAM gives the best model performance among different optimizers.

Objective

- To compare and analyze different adaptive learning rate techniques.

Dataset Description

- The weather forecast dataset is of Seattle Weather which includes different weather pattern and from those pattern rainfall pattern, sunny weather pattern, snowfall pattern are selected for the model.

- These weather pattern dataset have 4 features and 1 output with 2 classes.
- The cancer classification dataset have the basic classification of the cancer with 30 features and 1 output with 2 classes.

Sample Data of Seattle weather recording rainfall patterns

DATE	PRECIPITATION	TEMP_MAX	TEMP_MIN	WIND	RAIN
1/1/2012	0	12.8	5	4.7	0
1/2/2012	10.9	10.6	2.8	4.5	1
1/3/2012	0.8	11.7	7.2	2.3	1
1/4/2012	20.3	12.2	5.6	4.7	1
1/5/2012	1.3	8.9	2.8	6.1	1
1/6/2012	2.5	4.4	2.2	2.2	1
1/7/2012	0	7.2	2.8	2.3	1

Sample Data recording cancer patterns

RADIUS_MEAN	TEXTURE_MEAN	PERIMETER_MEAN	AREA_MEAN	DIAGNOSIS
17.99	10.38	122.8	1001	0
20.57	17.77	132.9	1326	0
19.69	21.25	130	1203	0
11.42	20.38	77.58	386.1	0
20.29	14.34	135.1	1297	0
12.45	15.7	82.57	477.1	0
18.25	19.98	119.6	1040	0

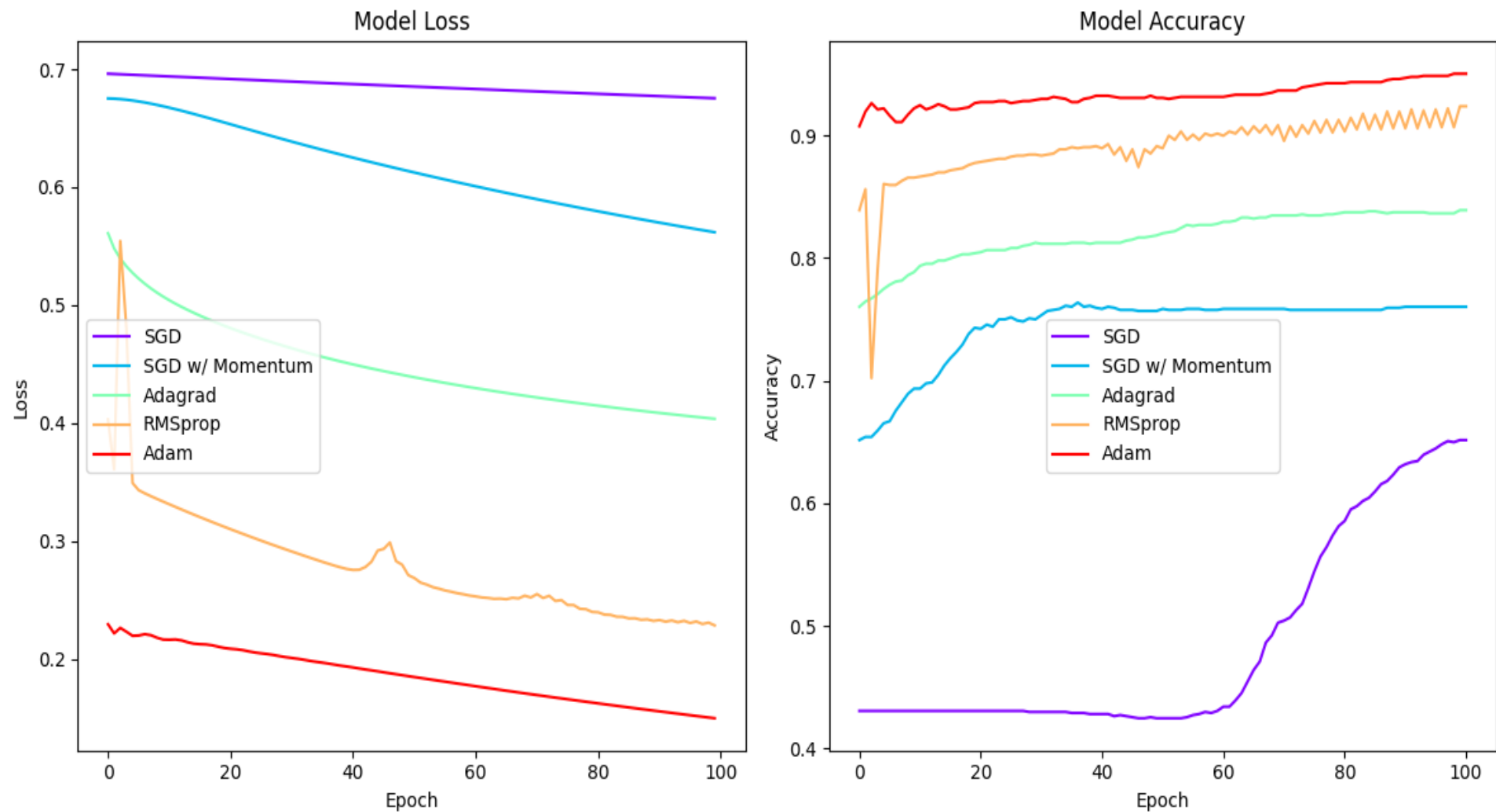
- sample of the processed weather dataset contains:
 - 1461 records of rainy weather pattern from 1st Jan 2012 to 31st Dec 2015 .
 - 5 feature variables and 1 target variable.
- sample of the processed cancer dataset contains:
 - 570 records of cancer pattern.
 - 30 feature variable and 1 target variable.

Sample table for Model Evaluation

OPTIMIZER	LEARNING RATE	AVERAGE TEST LOSS	TEST ACCURACY	MODEL TRAINING TIME (SEC)
SGD	0.001			
SGD with momentum(0.9)	0.001			
AdaGrad	0.001			
RMSProp	0.001			
ADAM	0.001			

Seattle weather rainfall data model

OPTIMIZER	LEARNING RATE	AVERAGE TEST LOSS	TEST ACCURACY	MODEL TRAINING TIME (SEC)
SGD	0.001	0.6755	0.6587	1.7426
SGD with momentum(0.9)	0.001	0.5709	0.7235	1.6318
AdaGrad	0.001	0.4135	0.8362	1.5572
RMSProp	0.001	0.2394	0.9044	1.6343
ADAM	0.001	0.2025	0.9386	1.6248



Seattle weather rainfall data model loss and accuracy graph

Seattle weather rainfall pattern dataset on the model fitted with different optimizer and similar learning rate shows:

- High test accuracy and low test loss for ADAM optimizer
- Less training time for AdaGrad but the accuracy is less than that of ADAM.

Cancer classification data model

OPTIMIZER	LEARNING RATE	AVERAGE TEST LOSS	TEST ACCURACY
SGD	0.001	0.5756	0.9556
SGD with momentum(0.9)	0.001	0.2114	0.9556
AdaGrad	0.001	0.1790	0.9556
RMSProp	0.001	0.2657	0.9556
ADAM	0.001	0.3864	0.9590

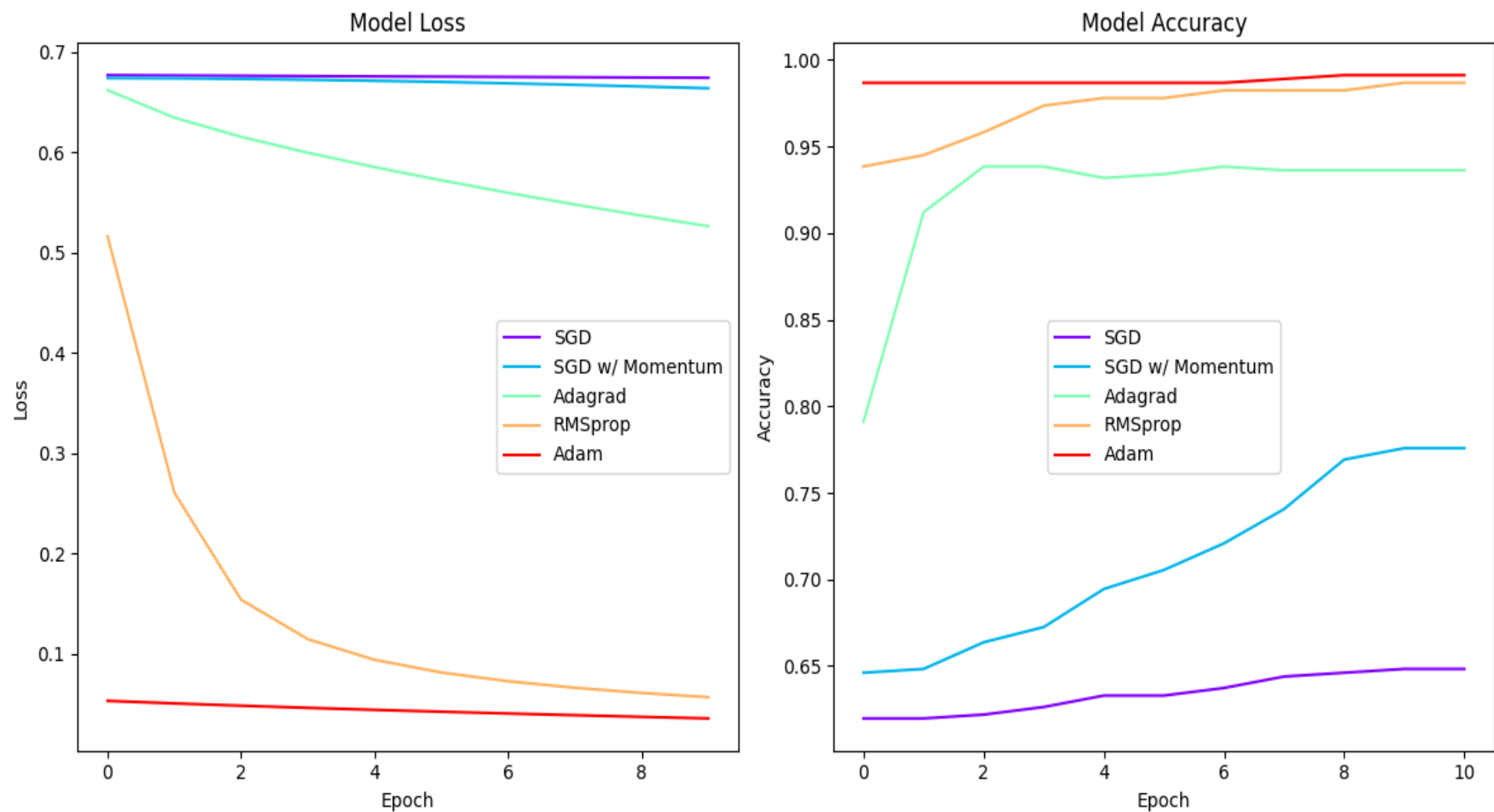


Figure 3.2 : Cancer classification data model loss and accuracy graph

Observation of table 3.2, Cancer classification pattern dataset on the model fitted with different optimizer and similar learning rate shows:

- High test accuracy for ADAM optimizer
- Less training time for RMSProp which has almost similar test accuracy of ADAM
- Less test loss for AdaGrad which also has similar test accuracy of RMSProp.

Table 3.3 : Seattle weather sunny data model

OPTIMIZER	LEARNING RATE	AVERAGE TEST LOSS	TEST ACCURACY
SGD	0.001	0.6703	0.6621
SGD with momentum(0.9)	0.001	0.5697	0.7304
AdaGrad	0.001	0.4202	0.8055
RMSProp	0.001	0.3529	0.8294
ADAM	0.001	0.3383	0.8396

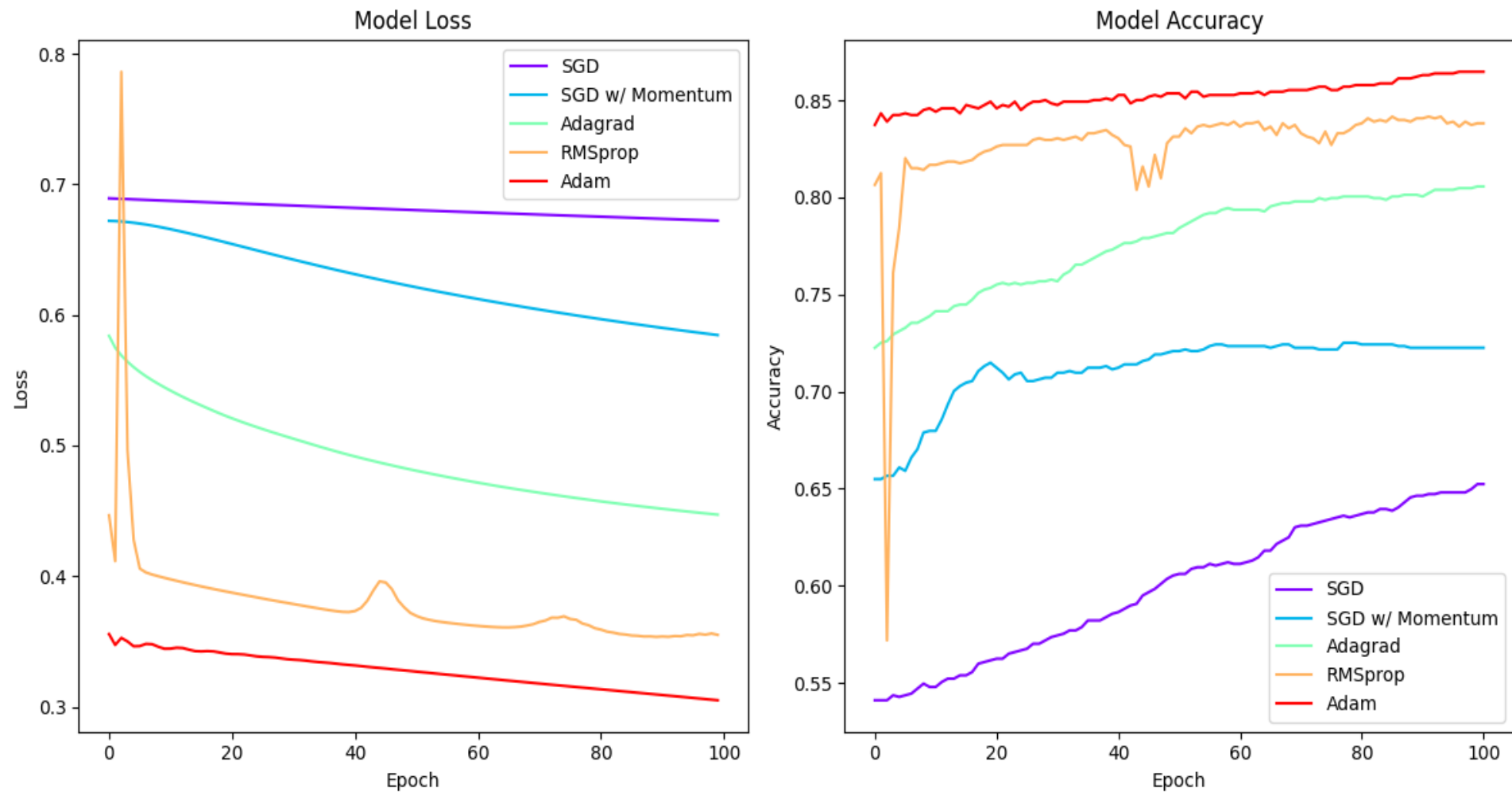


Figure 3.3 : Seattle weather sunny data model loss and accuracy graph

Table 3.4 : Seattle weather snowfall data model

OPTIMIZER	LEARNING RATE	AVERAGE TEST LOSS	TEST ACCURACY
SGD	0.001	0.4873	0.6863
SGD with momentum(0.9)	0.001	0.1282	0.7863
AdaGrad	0.001	0.0614	0.9363
RMSProp	0.001	0.0830	0.9761
ADAM	0.001	0.1356	0.9795

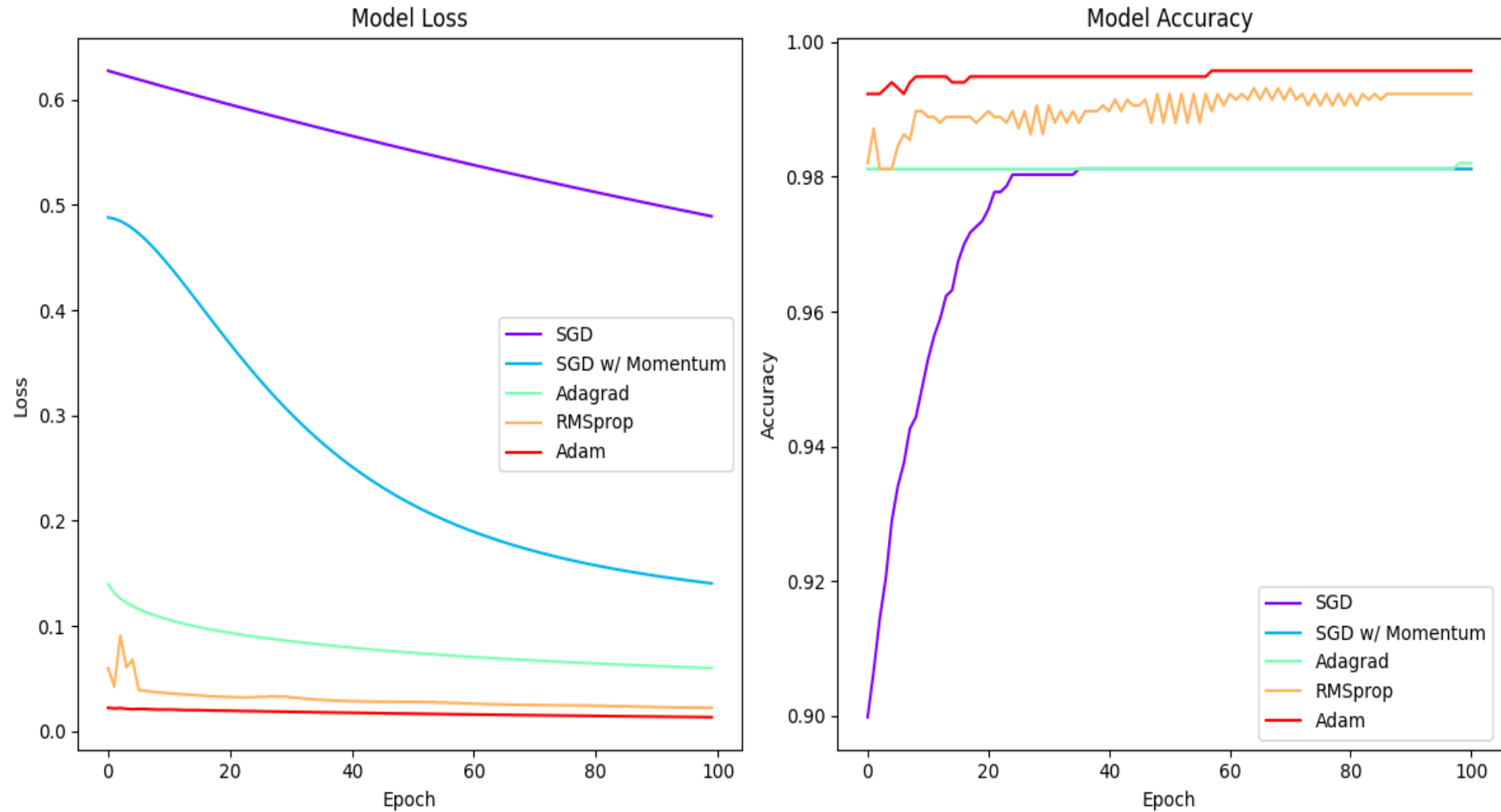


Figure 3.4 : Seattle weather snowfall data model loss and accuracy graph

- The ADAM optimizer have high model accuracy in comparison to other.
- Overall, from the test and the result it can be concluded that ADAM optimizer shows fast convergence speed and more test accuracy in comparison to other even though it may take more training time.
- The other optimizer like AdaGrad and RMSProp also shows high test accuracy similarly to ADAM optimizer and can be an alternative to ADAM optimizer.